

Network Engineering Project: Multi-Protocol Security Lab

1. Scenario & Objective

You are the Lead Network Engineer for GlobalCorp¹. You are validating a security policy between branch offices, a central Server Farm, and the Internet.

- **Branch A (EIGRP):** Contains the Finance Department (**PC1 & PC3**).
- **Branch B (RIP):** Contains the Sales Department (**PC2**)².
- **HQ (Core):** Routers R1 & R2 act as the transit backbone (OSPF & HSRP)³.
- **HQ (Server Farm):** R2 now hosts the corporate Server Farm (Network 20.0).

The Mission:

1. **Connectivity:** All PCs must route traffic to the Internet (ISP) via the HQ backbone⁴.
2. **Routing Integration:** Connect EIGRP, RIP, and OSPF without using Route Redistribution ⁵, using "Static Route Glue" instead⁶.
3. **Security Constraint:** Implement **two specific ACLs** to secure the Finance Department and the Server Farm.

2. Topology Design

- **R3 (Branch A):** Connected to **PC1 & PC3** (Simulating the Finance Network)⁷.
- **R4 (Branch B):** Connected to **PC2** (Simulating the Sales Network)⁸.
- **R1 & R2 (HQ):** Connected via a switch to form the Core⁹.
- **R2 (Server Link):** Connected to the **Server Farm** (Server 1 & Admin PC).
- **R5 (ISP):** Connected to R1¹⁰.

3. IP Addressing Table

Device	Interface	IP Address	Subnet Mask	Gateway	Description
R1 (HQ)	Gi0/1	DHCP	-	-	ISP Uplink ¹¹
	Gi0/0	192.168.1.2	/24	-	Core Link (Active) ¹²

Device	Interface	IP Address	Subnet Mask	Gateway	Description
	Se0/0/0	10.1.1.1	/30	-	Link to Branch A ¹³
R2 (HQ)	Gi0/0	192.168.1.3	/24	-	Core Link (Standby) ¹⁴
	Se0/0/0	10.2.2.1	/30	-	Link to Branch B ¹⁵
	Gi0/1	192.168.20.1	/24	-	Server Gateway
R3 (Br A)	Se0/0/0	10.1.1.2	/30	-	Link to HQ ¹⁶
	Gi0/0	192.168.3.1	/24	-	Link to Finance ¹⁷
R4 (Br B)	Se0/0/0	10.2.2.2	/30	-	Link to HQ ¹⁸
	Gi0/0	192.168.4.1	/24	-	Link to Sales ¹⁹
PC1	NIC	192.168.3.10	/24	192.168.3.1	Finance Station 1 ²⁰
PC3	NIC	192.168.3.11	/24	192.168.3.1	Finance Station 2
PC2	NIC	192.168.4.10	/24	192.168.4.1	Sales Station ²¹
Server	NIC	192.168.20.100	/24	192.168.20.1	Corp Server
Admin PC	NIC	192.168.20.10	/24	192.168.20.1	IT Admin
R5 (ISP)	Gi0/1	203.0.113.1	/24	-	ISP Gateway ²²

4. Configuration Tasks

Task 1: End-Device Setup

- **Configure PC1:** IP 192.168.3.10, Gateway 192.168.3.1²³.
- **Configure PC3:** IP 192.168.3.11, Gateway 192.168.3.1.
- **Configure PC2:** IP 192.168.4.10, Gateway 192.168.4.1²⁴.
- **Configure Server:** IP 192.168.20.100, Gateway 192.168.20.1.
- **Configure Admin PC:** IP 192.168.20.10, Gateway 192.168.20.1.

Task 2: Core & ISP Setup (OSPF, HSRP, DHCP)

- **ISP (R5):** Configure as DHCP Server for 203.0.113.0/24²⁵.
- **HQ (R1):** Configure Gi0/1 as ip address dhcp²⁶.
- **HQ Core (R1 & R2):**
 - Enable **OSPF Area 0** advertising 192.168.1.0/24²⁷.
 - Enable **HSRP** on Gi0/0. VIP: 192.168.1.1. Set R1 Priority to 150 (Active)²⁸.

Task 3: Branch Protocols (EIGRP & RIP)

- **Branch A (R3):** Enable **EIGRP 100**. Advertise the link to HQ (10.1.1.0) and the Finance network (192.168.3.0)²⁹.
- **Branch B (R4):** Enable **RIPv2**. Advertise the link to HQ (10.2.2.0) and the Sales network (192.168.4.0)³⁰.

Task 4: The "Static Glue" (Routing Logic)

Since Redistribution is banned, we must manually tell HQ where the PCs are³¹.

- **On R1 (HQ Active):**
 - Create a Static Route to the Sales Network (PC2) pointing to R2³².
 - Create a Default Route to the ISP³³.
- **On R2 (HQ Standby):**
 - Create a Static Route to the Finance Network (PC1/PC3) pointing to R1³⁴.
- **On R3 & R4:**
 - Create a Default Route pointing to their respective HQ routers³⁵.

Task 5: Security Policy (Dual ACLs)

ACL 1: Finance Department Protection

- **Objective:** Prevent Sales (192.168.4.0) from accessing the Finance pc1 (192.168.3.1), but allow to the subnet 192.168.3.0 and to Internet and Server access.

ACL 2: Server Farm Lockdown

- **Objective:** Protect the Server Farm (192.168.20.0).

- Allow **Admin PC** full access (local).
- Allow **sales** full access.
- Block **pc1 from finance** from Web (HTTP/80) access on the server. allow everything else (including Ping).

5. Verification Checklist

- **[] Routing & Connectivity:**
 - From **PC1**, ping **PC3** (Local LAN Check).
 - From **PC3**, ping **203.0.113.1** (ISP Check).
 - From **PC2**, ping **203.0.113.1** (ISP Check).
 - From **Admin PC**, ping **203.0.113.1** (ISP Check).
 - From **Admin PC**, ping **PC1** (Core Routing Check).
- **[] Routing Glue Check:**
 - From **PC2**, ping **PC3**.
 - From Admin pc ping pc3.
 - *Result: **Success.** (Proves Static Route "Glue" works).*
- **[] check HSRP BY SHOW STANDBY COMMAND ON R1 AND R2.**
- **[] Security Check - ACL 1 (Finance):**
 - From **PC2**, ping **PC1**. Result: **Destination Unreachable** (Blocked).
 - From **PC2**, ping **PC3**. Result: **success.**
- **[] Security Check - ACL 2 (Server):**
 - From **PC1**, ping **Server (192.168.20.100)**. Result: **success**
 - From **PC1**, use Web Browser to access **192.168.20.100**. Result: **blocked.**
 - From **PC3**, use Web Browser to access **192.168.20.100**Result: **Success** (Allowed).